Problem Set 4

Submission:
Friday, 13/03/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.
Only problems marked with $(*)$ will be graded.

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1. **$(*)$ Benford's law and σ-algebras** [4 Points]

   Empirical evidence suggests that the first digits in lists of large data sets of real numbers appearing naturally (such as natural constants, lengths of rivers, stock prices, ...) are *not* uniformly distributed on $\{1, \ldots, 9\}$. Instead, they follow the so-called Benford distribution, which is a discrete distribution on $\{1, \ldots, 9\}$ with
$$p(k) = \log_{10}\left(1 + \frac{1}{k}\right), \qquad k \in \{1, \ldots, 9\}$$
   (Newcomb 1881, Benford 1938). We want to make this mathematically plausible, and aim at constructing a probability measure $\mathbf{P}$ on $(0, \infty)$ equipped with an appropriate σ-algebra $\mathscr{M}$ with
$$(\star) \qquad \mathbf{P}[\{\omega \in (0,\infty) \,:\, \text{the decimal representation of } \omega \text{ starts with digit } k\}] = \log_{10}\left(1 + \frac{1}{k}\right).$$
   It is clear that $\mathbf{P}$ has to be scale invariant for all $\alpha > 0$, i.e. $\mathbf{P}[\alpha B] = \mathbf{P}[B]$ for all $\alpha > 0$ and $B \in \mathscr{M}$.[1] One can show that for $\mathscr{B}((0,\infty)) = \{B \cap (0,\infty) \,:\, B \in \mathscr{B}(\mathbb{R})\}$, no such probability measure exists. However, one can consider a smaller σ-algebra
$$\mathscr{M} = \sigma(\mathscr{E}), \text{ where } \mathscr{E} = \{M_t \,:\, 1 < t < 10\}, \text{ with sets } M_t = \bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, t).$$
   One can then show that the expression $\mathbf{P}[M_t] := \log_{10}(t)$ can be extended to a probability measure $\mathbf{P}$ on $((0,\infty), \mathscr{M})$ (you do not need to show this).

   (a) Show that $\alpha M_t \in \mathscr{M}$ for all $\alpha > 0$ and all $M_t \in \mathscr{E}$.
   (b) Show that $\mathbf{P}[\alpha M_t] = \mathbf{P}[M_t]$.
   (c) Show that $(\star)$ holds.
   (d) Show that the properties of (a) and (b) hold for all $M \in \mathscr{M}$.
   Hint: Show that $\mathscr{M}' = \{M \in \mathscr{M} \,:\, \alpha M \in \mathscr{M}\}$ is a σ-algebra. You can use without proof that $\mathscr{M}'' = \{M \in \mathscr{M} \,:\, \mathbf{P}[\alpha M] = \mathbf{P}[M]\}$ also a σ-algebra. Why does this imply the result?

2. **More on σ-algebras**

   (a) Give an example showing that the union $\mathscr{F} \cup \mathscr{G}$ of two σ-algebras $\mathscr{F}$ and $\mathscr{G}$ on a sample space $\Omega \neq \varnothing$ is not necessarily a σ-algebra on Ω.
   (b) Recall that the Borel-σ-algebra $\mathscr{B}(\mathbb{R})$ is defined as
$$\mathscr{B}(\mathbb{R}) = \sigma\big(\{[a,b)\,;\, a,b \in \mathbb{R}, a < b\}\big).$$
   Show that for every $a, b \in \mathbb{R}$, one has $[a,b] \in \mathscr{B}(\mathbb{R})$ and $(-\infty, a] \in \mathscr{B}(\mathbb{R})$.

[1] This is since the quantity of interest is, say, a length, and it should be invariant under changing the unit of measurement.

(c) Show that the Borel-σ-algebra $\mathscr{B}(\mathbb{R})$ is also generated by the collection of all sub-intervals of the form $(-\infty, b]$ and the collection of all open sets in $\mathbb{R}$.

Hint: Define $\mathscr{B}_1 = \sigma(\{O \subseteq \mathbb{R}\,;\, O \text{ is open}\})$ and $\mathscr{B}_2 = \sigma(\{(-\infty, b]\,;\, b \in \mathbb{R}\})$. Then show that $\mathscr{B}(\mathbb{R}) \subseteq \mathscr{B}_2 \subseteq \mathscr{B}_1$, and finally $\mathscr{B}_1 \subseteq \mathscr{B}(\mathbb{R})$. You can use the fact that every open subset O of $\mathbb{R}$ can be written as a countable union of disjoint open intervals in the last step.

3. **Calculations with continuous distributions**

(a) Suppose the life-time of a lightbulb is exponentially distributed with parameter $\lambda > 0$, where time is measured in years.

i. What is the probability that the lightbulb lasts for at least 2, but at most 3 years? Give a formula for general λ and then specify it to $\lambda = 1\frac{1}{\text{Year}}$.

ii. For general $\lambda > 0$, calculate the half-life, that is the value $t > 0$ such that the probability for the lightbulb to last at least until time t is exactly $\frac{1}{2}$.

(b) Consider again the exponential distribution $\mathbf{P} = \mathscr{E}(\lambda)$. Show that it satisfies the memoryless property, i.e. show that for $s, t > 0$, one has

$$\mathbf{P}\big[(s+t, \infty) | (s, \infty)\big] = \mathbf{P}\big[(t, \infty)\big].$$

(c) The *continuous Laplace[2]-distribution with parameter* $\lambda > 0$ is a probabiltiy measure $\mathbf{P}$ on $\mathbb{R}$ characterized by the density

$$f(x) = c_\lambda \exp\left(-\frac{|x|}{\lambda}\right), \qquad x \in \mathbb{R}.$$

i. Determine the constant c_λ.

ii. Determine for the cumulative distribution function $F(x)$ for $x \in \mathbb{R}$.

iii. Calculate $\mathbf{P}[[-1, 1]]$.

4. **($*$) On probability density functions** [4 Points]

(a) Let

$$f(x) := \frac{1}{x^k} \mathbb{1}_{[1,+\infty)}(x).$$

For what value of $k \in \mathbb{R}$, if any, is f a probability density function?

(b) Give an example of a probability density function f such that $c\sqrt{f}$ cannot be a probability density function for any $c > 0$.

(c) Let

$$f(x) = 2|x| \left(1 - x^2\right) \mathbb{1}_{\{|x| \leq 1\}}.$$

i. Find the corresponding cumulative distribution function F.

ii. Let $\mathbf{P}$ be the associated probability measure. Compute $\mathbf{P}[(-\infty, -\frac{1}{2})]$ and $\mathbf{P}[[-\frac{1}{2}, \frac{1}{2}]]$.

[2] This should not be confused with "Laplace probability space".